

4 A rocket is midway between the Earth and Moon as shown in Fig. 4.1.

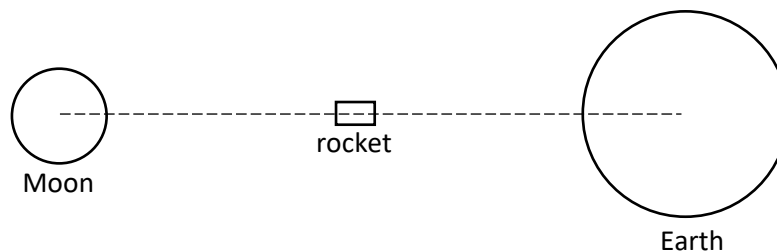


Fig. 4.1

The distance between the centers of mass of the Moon and Earth is 384×10^3 km. The mass of the Moon is 7.35×10^{22} kg and the mass of the Earth is 5.97×10^{24} kg.

(a)	Calculate the escape velocity of the rocket when it is midway between the Earth and Moon.
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escape speed = m s^{-1} [3]

The gravitational potential energy of the rocket = $-\frac{GM_E m}{R_E} - \frac{GM_M m}{R_M}$

Where M_E is the mass of the earth, M_M the mass of the Moon, R_E the distance between the center of mass of the Earth and the rocket and R_M the distance between the center of mass of the Moon and the rocket.

Hence the gravitational potential energy = $-m(\frac{GM_E}{R_E} + \frac{GM_M}{R_M})$

$$= -m(\frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{384 \times 10^6 / 2} + \frac{6.67 \times 10^{-11} \times 7.35 \times 10^{22}}{384 \times 10^6 / 2})$$

$$= -m(2.099 \times 10^6) J \text{ [C1]}$$

The minimum kinetic energy required to escape is the gain in gravitational potential energy needed to enable the rocket to escape to infinity, hence

$$\frac{1}{2}mv_e^2 = m(2.099 \times 10^6) \text{ [C1]}$$

$$v_e = 2050 \text{ m s}^{-1} \text{ [A1]}$$

(b)	Explain qualitatively how the escape velocity of the rocket will change as its position is moved nearer the moon from its current position.
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5	Two small coherent sound sources S_1 and S_2 are set up as shown in Fig 5.1 below
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